



Effects of replacing inorganic salts of trace minerals with organic trace minerals in pre- and postpartum diets on feeding behavior, rumen fermentation, and performance of dairy cows

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ABSTRACT

Our objectives were to evaluate the effects of complete replacement of supplementary inorganic salts of trace minerals (STM) by organic trace minerals (OTM) in both pre- and postpartum diets on feeding behavior, ruminal fermentation, rumination activity, energy metabolism, and lactation performance in dairy cows. Pregnant cows and heifers ($n = 273$) were blocked by parity and body condition score and randomly assigned to either STM or OTM diets at 45 ± 3 d before their expected calving date. Both groups received the same diet, except for the source of trace minerals (TM). The STM group was supplemented with Co, Cu, Mn, and Zn sulfates and Na selenite, whereas the OTM group was supplemented with Co, Cu, Mn, and Zn proteinates and selenized yeast. Treatments continued until 156 days in milk and pre- and postpartum diets were formulated to meet 100% of recommended levels of each TM in both treatments, taking into consideration both basal and supplemental levels. Automatic feed bins were used to assign treatments to individual cows and to measure feed intake and feeding behavior. Rumination activity was monitored by sensors attached to a collar from wk -3 to 3 relative to calving. Blood metabolites were evaluated on d -21 , -10 , -3 , 0 , 3 , 7 , 10 , 14 , 23 , and 65 relative to calving. Ruminal fluid samples were collected using an ororuminal sampling device on d -21 , 23 , and 65 relative to calving, for measurement of ruminal pH and concentration of volatile fatty acids. Cows were milked twice a day and milk components were measured monthly. Cows supplemented with OTM tended to have longer daily feeding time (188 vs. 197 min/d), and

greater dry matter intake (DMI; 12.9 vs. 13.3 kg), and had a more positive energy balance (3.6 vs. 4.2 Mcal/d) and shorter rumination time per kg of dry matter (DM; 40.1 vs. 37.5 min/kg of DM) than cows supplemented with STM during the prepartum period. In the postpartum period, OTM increased DMI in multiparous cows (24.1 vs. 24.7 kg/d) but not in primiparous cows (19.1 vs. 18.7 kg/d). The difference in DMI of multiparous cows was more evident in the first 5 wk of lactation, when it averaged 1 kg/d. Milk yield was not affected by treatment in multiparous cows (44.1 vs. 44.2 kg/d); however, primiparous cows supplemented with OTM had lesser yields than primiparous cows supplemented with STM (31.9 vs. 29.8 kg/d). Cows supplemented with OTM had a greater percentage of protein in milk (3.11 vs. 3.17%), reduced concentration of nonesterified fatty acids in serum (0.45 vs. 0.40 mmol/L), and rumination activity (30.1 vs. 27.8 min/kg of DM) than cows supplemented with STM. At the end of the transition period, cows supplemented with OTM had reduced molar proportion of acetate, reduced pH, and tended to have a greater concentration of total volatile fatty acids in ruminal fluid. In conclusion, complete replacement of STM by OTM caused modest changes in rumen fermentation, feeding behavior, energy metabolism, and performance of dairy cows, improving postpartum DMI in multiparous cows and reducing circulating levels of nonesterified fatty acids. The pre-absorptive effects of TM source and the parity specific responses on performance warrant further research.

Key words: organic trace mineral, nutrition, feeding behavior, transition period, milk yield

INTRODUCTION

Trace minerals (TM) represent a small but important portion of a dairy cow diet. They serve as cofactors of metalloenzymes and are required for several biological processes, including regulation of oxidative

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balance, synthesis of vitamins and proteins, and immune cell function (Goff, 2018). Because of insufficient concentration in most common feeds, dietary supplementation of some TM (e.g., Co, Cu, Mn, Se, and Zn) is recommended to optimize health and performance (NRC, 2001). Inorganic salts of trace minerals (**STM**) such as sulfates, oxides, carbonates, and chlorides are widely available and represent an inexpensive form of supplementation. However, the ionic bond in STM often dissociates during passage through the gastrointestinal tract (**GIT**), allowing interactions with other molecules, which reduces bioavailability of TM for absorption (Spears, 1996, 2003).

Two alternatives to STM supplements are the hydroxy trace minerals (**HTM**), another inorganic form of TM but with covalent bonds between the TM and hydroxyl groups, and the organic sources of trace minerals (**OTM**), which are chelates between the TM and an organic molecule such as a peptide, AA, polysaccharide, propionate, acetate, or picolinate (Goff, 2018). The covalent bonds and the ring-like structure of these alternatives are believed to protect the TM from most unwanted chemical reactions in the GIT, especially in the rumen, increasing bioavailability of TM for intestinal absorption (Spears, 1996, 2003). The claimed improvements would enhance TM availability in the vascular system and tissues, which could improve enzymatic and cellular processes that ultimately benefit health and performance (Spears and Weiss, 2008).

Sources of supplementary TM with different stability, solubility, and chemical properties could also influence the biology of microorganisms in the GIT, potentially causing pre-absorptive effects. Trace minerals are required for proper rumen microbial function (Durand and Kawashima, 1980); however, excessive amounts of some TM have toxic effects to rumen microbes and impair fiber digestibility (Hubbert et al., 1958; Martinez and Church, 1970). Thus, excessive dissociation of STM in the rumen can negatively affect rumen digestibility and fermentation with possible consequences to performance (Genther and Hansen, 2015; Faulkner and Weiss, 2017). Supplementation with OTM seemed to provide adequate amounts of TM to rumen microbe utilization, favoring fiber digestibility and rumen fermentation (Galbraith et al., 2016; Pino and Heinrichs, 2016; Guimaraes et al., 2020), which could also affect passage rate, rumination activity, and feeding behavior of cows (Miller et al. 2020a). Such pre-absorptive effects could help explain differences in performance caused by sources of supplementary TM that could not otherwise be explained by the measures of postabsorptive mineral status of cows.

Multiple researchers have evaluated the effects of supplementing OTM on dairy cow performance; how-

ever, the diverse range of experimental designs (e.g., adding OTM on top of existing mineral premix, partial or complete replacement of STM by OTM) and the variable results make the comparison with STM inconclusive. Moreover, the extrapolation of research results with a specific product or products must be done carefully because of the different formulations, manufacturing technology, and chemical features of supplements (Byrne et al., 2021). Nevertheless, positive results have been reported with partial replacement of STM (Co, Cu, Mn, Zn) by OTM during the transition period (Siciliano-Jones et al., 2008; Osorio et al., 2016). Although partial replacement is a common practice in the industry, the interpretation of studies using partial replacement is challenging because of the confounding effects of source, inclusion levels, and their combination or interaction. Studies evaluating the complete replacement of STM by OTM are easier to interpret, but scarcer in the literature. Yasui et al. (2019) reported no difference in performance when mid-lactation cows were fed STM (Cu, Mn, Zn) or OTM for 6 wk. In contrast, Nocek et al. (2006) reported greater milk yield when replacing STM (Co Cu, Mn, Zn) by OTM for 2 lactation cycles. None of the studies above included Se as part of the experimental treatments. In fact, inorganic (Na selenite or selenate) and organic (selenized yeast or selenomethionine) sources of Se are typically compared separately, and results of performance are also variable and those related to digestive physiology are scarce (Weiss and Hogan, 2005; Silvestre et al., 2007; Rutigliano et al., 2008; Calamari et al., 2010; Galbraith et al., 2016).

Our objectives were to evaluate the effect of complete replacement of supplementary STM (sulfates and Na selenite) by OTM (proteinates and selenized yeast) in both pre- and postpartum diets, fed to meet 100% of recommended levels in the diet when considering both basal and supplemental levels, on feeding behavior, rumen fermentation, and lactation performance of dairy cows. We hypothesized that total replacement of STM by OTM in pre- and postpartum diets would improve rumen function and TM status in dairy cows, ultimately benefiting performance.

MATERIALS AND METHODS

Research methods were approved by the University of Guelph Animal Care Committee (Animal Utilization Protocol #4064).

Animals, Housing, Diets, and Experimental Design

The experiment was performed at the Ontario Dairy Research Centre (**ODRC**; Elora, ON, Canada) between

December 2018 and September 2020. A preparation phase started 8 mo earlier, in April 2018, with selection of TM sources and adjustment of their inclusion levels in the diets of pregnant heifers and pregnant cows to avoid TM buildup in tissues and possible carryover effects during the experimental period. Briefly, a retrospective analysis of TM concentration in feed ingredients sampled monthly over a period of 2 yr (2016–2018) was performed to estimate the basal levels of TM in the TMR without mineral supplements. Samples were collected by ODRC staff and analyses were performed at the A & L Canada Laboratories (London, ON, Canada) using partial digestion in nitric and hydrochloric acid at 1:3 ratio and inductively coupled plasma atomic emission spectroscopy (Thermo Scientific iCAP 6000, Thermo Scientific). The supplementation of TM was then adjusted to meet 100% of recommended levels in the TMR (basal + supplement) and only STM were fed. Inclusion levels were based on scientific recommendations available at the time of development of the experimental design, and attempted to anticipate the new guidelines to be published in NASEM (2021). Basal levels were considered to calculate the inclusion levels of Cu, Mn, and Zn but were ignored for Co and Se, due to their lower concentrations in feed ingredients and imprecise measurements. The average basal concentrations of Cu, Mn, and Zn in the prepartum TMR without TM supplements were 5.9, 26.1, and 22.7 mg/kg, respectively. The average basal concentrations of Cu, Mn, and Zn in the postpartum TMR without TM supplements were 6.5, 22.0, and 25.6 mg/kg, respectively.

Two hundred eighty-two pregnant Holstein dairy cows and heifers were enrolled 45 ± 3 d before the expected calving date in small weekly groups (2–8 cows) between December 2018 and February 2020. At enrollment, cows were blocked by parity and BCS and randomly assigned to receive 1 of 2 dietary treatments: STM or OTM. Animals remained in the assigned treatment until the end of the experiment at 156 DIM. Cows enrolled in both treatments received the same basal TMR during the prepartum period and the same basal TMR during the postpartum period, except for the source of supplementary TM (Table 1). Cows enrolled in the STM diets were supplemented with Co, Cu, Mn, and Zn sulfates and Na selenite. Cows enrolled in the OTM diets were supplemented with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech). A nonselenized yeast product (Diamond V Yeast XP, Diamond V) was added to both treatments to avoid a confounding factor by the presence of *Saccharomyces cerevisiae* in only one of the treatments. The inclusion levels of supplementary TM were also the same for both treatments and were targeted to

provide 0.25 mg/kg of Co, 13.7 mg/kg of Cu, 40.0 mg/kg of Mn, 0.3 mg/kg of Se, and 40.0 mg/kg of Zn in the prepartum diets and 0.25 mg/kg of Co, 15.7 mg/kg of Cu, 40.0 mg/kg of Mn, 0.3 mg/kg of Se, and 63.0 mg/kg of Zn in the postpartum diets. Similar to the preparation phase, basal supply of TM in feed ingredients was considered for the inclusion levels of Cu, Mn, and Zn, but not for the inclusion levels of Co and Se. After discounting the estimated basal levels (established in the preparation phase) from the final target levels, the supplementary contributions of Cu, Mn, and Zn were 7.8, 13.9, and 17.3 mg/kg, respectively, in the prepartum TMR and 9.2, 18.0, and 37.4 mg/kg, respectively, in the postpartum TMR.

To manage feeding treatments on the research farm, 4 TM premixes were created: prepartum inorganic, prepartum organic, postpartum inorganic, and postpartum organic. Trace mineral premixes were then incorporated into protein pellet supplements that were added to the various TMR. Prepartum protein pellets of both treatments were identical, except for the TM premix. Postpartum protein pellets of both treatments were identical, except for the TM premix. Premixes and pellet supplements were prepared by Floradale Feed Mill Ltd., which is certified by the Canadian Food Inspection Agency (CFIA) and Hazard Analysis Critical Control Point (HACCP). The composition of the diets, including the pellet supplements, is described in Table 1. Every morning, forages and high moisture corn were mixed in a mixer wagon (Haybuster H-1150) at the feed center, brought to the cow barn, and distributed into horizontal mixers (V-Rotor 6100, Valmetal), where the protein pellet supplement was added to the TMR and mixed again for 8 min. From the horizontal mixers, the diets were distributed to assigned feed bins.

Prepartum cows were housed in 3 freestall pens, equipped with automatic feed bins (Insentec B. V.) that measure feed intake and feeding behavior (as validated by Chapinal et al., 2007), and mattress-beds with chopped wheat straw bedding. Postpartum cows were housed in 4 freestall pens, equipped with automatic feed bins, and mattress-beds with chopped wheat straw bedding. The number of cows in each pen and the number of cows of each treatment per pen were maintained similar throughout the experiment. Prepartum diets were delivered once daily at 1030 h and postpartum diets were delivered in 2 parts, 60% at 0900 h and 40% at 1530 h, using feed carts (Supercart VA-542, Valmetal). Refusals were removed from the feed bins daily, approximately 30 min before the morning feeding, and the amount of feed offered was adjusted daily to allow 8% refusals.

Two days before the expected calving date, or when demonstrating signs of calving, cows were moved

Table 1. Feed ingredients and chemical composition (mean $\pm$ SD) of dry cow and lactating cow TMR¹

Item	Dry cow TMR	Lactating cow TMR
Ingredient, % of DM		
Corn silage	39.1 $\pm$ 0.2	29.4 $\pm$ 0.3
Alfalfa haylage	5.9 $\pm$ 0.2	29.6 $\pm$ 0.3
Wheat straw	32.1 $\pm$ 0.2	1.8 $\pm$ 0.02
High-moisture corn	—	25.0 $\pm$ 0.4
Dry cow supplement ²	23.0 $\pm$ 0.1	—
Lactating cow supplement ³	—	14.2 $\pm$ 0.03
DM, %	54.5 $\pm$ 2.6	47.3 $\pm$ 2.0
Chemical composition, % of DM		
CP	12.19 $\pm$ 0.44	15.72 $\pm$ 0.38
NFC	28.20 $\pm$ 0.91	45.07 $\pm$ 0.80
Starch	14.82 $\pm$ 1.23	27.08 $\pm$ 0.91
ADF	31.52 $\pm$ 1.07	19.35 $\pm$ 0.83
NDF	48.83 $\pm$ 1.27	29.28 $\pm$ 1.08
TDN	63.99 $\pm$ 1.77	73.45 $\pm$ 1.64
Crude fat	2.75 $\pm$ 0.33	3.63 $\pm$ 0.35
Ash	7.07 $\pm$ 1.07	6.56 $\pm$ 0.41
Ca	1.41 $\pm$ 0.02	0.82 $\pm$ 0.02
P	0.29 $\pm$ 0.03	0.41 $\pm$ 0.01
Mg	0.45 $\pm$ 0.02	0.36 $\pm$ 0.01
K	1.11 $\pm$ 0.13	1.40 $\pm$ 0.13
Na	0.11 $\pm$ 0.01	0.34 $\pm$ 0.01
S	0.40 $\pm$ 0.01	0.21 $\pm$ 0.01
Cl	0.85 $\pm$ 0.07	0.55 $\pm$ 0.10
DCAD, mEq/kg	-153.48 $\pm$ 30.04	218.44 $\pm$ 43.17
NE _L , Mcal/kg	1.45 $\pm$ 0.04	1.68 $\pm$ 0.04
Trace mineral, mg/kg of DM		
Co	0.37 $\pm$ 0.07	0.29 $\pm$ 0.06
Cu	12.57 $\pm$ 1.41	13.39 $\pm$ 0.98
Fe	318.44 $\pm$ 114.39	340.00 $\pm$ 111.96
Mn	56.97 $\pm$ 6.20	47.63 $\pm$ 6.34
Mo	1.33 $\pm$ 0.19	1.58 $\pm$ 0.26
Se	0.43 $\pm$ 0.08	0.36 $\pm$ 0.04
Zn	51.14 $\pm$ 7.16	59.68 $\pm$ 4.53

¹Chemical compositions of the TMR and individual feed ingredients were evaluated once a month throughout the study, and small adjustments in diet formulations were made by dairy nutritionists based on estimated MP and ME.

²Dry cow pellet supplement supplied by Floradale Feed Mill Ltd. was composed of (DM basis) 23.55% soybean meal 47, 12.36% Animate (Phibro), 13.2% beet pulp, 10.41% ground soy hulls, 9.8% high-bypass soybean meal, 9.67% limestone calcium carbonate, 6.00% bran, 4.72% canola, 2.52% molasses, 1.77% calcium sulfate, 1.49% Diamond V Yeast XP (Diamond V), 1.04% magnesium oxide, 1.02% inorganic or organic premix according to experimental design, 0.76% Integral (Alltech), 0.82% fine salt, 0.66% monocalcium phosphate, 0.17% granular sulfur, and 0.04% Rumensin/Coban (Elanco).

³Lactating cow pellet supplement supplied by Floradale Feed Mill Ltd. was composed of (DM basis) 44.02% SoyPlus (Landus Cooperative), 14.7% soybean meal 47, 10.19% wheat shorts, 9.83% canola, 3.39% sodium sesquicarbonate, 3.33% limestone Ca carbonate, 3.39% fine salt, 2.77% monocalcium phosphate, 1.97% magnesium oxide, 1.49% Diamond V Yeast XP (Diamond V), 1.11% tallow, 1.10% DCAD+, 0.97% MetaSmart (Adisseo), 0.76% Integral (Alltech), 0.74% inorganic or organic premix according to experimental design, 0.32% granular sulfur, and 0.04% Rumensin/Coban (Elanco).

from the prepartum pens to individual box stalls with chopped wheat straw bedding and remained there until 7 DIM when they were moved to postpartum freestall pens. The individual stalls were also used for cows diagnosed with a clinical disease that required treatment. For the individual stalls, diets were delivered using a feed cart equipped with an electronic scale (Super Data Ranger, American Calan) and refusals were weighed daily. Daily feed intake in the individual stalls was calculated as the weight difference between offered feed and refusals.

During the entire prepartum period and during the early postpartum period, each cow was assigned to one individual feed bin (a 1:1 cow to feed bin ratio). Approximately 6 wk after calving, cows were moved to a new lactating cow pen and allocated in a 3:2 cow to feed bin ratio to accommodate the space available for the herd as a whole and the number of feed bins available. Each group of 3 cows sharing feed bins was formed within parity and treatment groups, and the 2 bins shared by a group of 3 cows were always located beside each other. Each automatic feed bin allowed

only one cow to eat at a time, so individual feed intake and individual feeding behavior were recorded based on individual radio frequency identification tags.

Analysis of Feed Samples

Samples of TMR and individual feed ingredients were collected monthly by ODRC staff for analysis of chemical composition by wet chemistry at the A & L Canada Laboratories (London, ON, Canada). Diet formulations were evaluated monthly by the dairy nutritionist and small adjustments in the formulation were made based on estimated metabolizable protein and energy. In addition, samples of TMR were collected weekly by the research team to form composite monthly samples for analyses of TM concentration. Samples were placed in metal trays, weighed, and oven-dried at 60°C for 48 h. After drying, feed samples were weighed again for determination of DM and digested in a solution of nitric acid, hydrochloric acid, and standards using a Teflon vessels system (MARexpress, CEM) in a microwave digestion protocol. The microwave-digested sample was resuspended in Nanopure water. The clear extract supernatant was further diluted by the ESI prepFAST Automated Inline Dilution System (Elemental Scientific) by a factor of 10 and analyzed by inductively coupled plasma mass spectrometry (Agilent 7900 ICPMS, Agilent).

Milk Yield and Composition

Cows were milked twice a day at 0530 and 1700 h in a rotary parlor (DeLaval). Milk yields were recorded automatically at each milking. Milk samples for composition analysis were collected once a month. Morning and afternoon milking samples were mixed and fat, protein, and SCC were analyzed by Fourier-Transform Infrared spectroscopy (Milkoscan FT+ and Milkoscan 6000, Foss) at a DHIA testing laboratory (Lactanet, Guelph, ON, Canada). Energy-corrected milk was calculated by the following equation: $ECM = [(0.327 \times \text{kg of milk}) + (12.95 \times \text{milk fat}) + (7.20 \times \text{milk protein})]$.

BW, BCS, and Energy Balance

Body weight was measured at the enrollment, once a week during the prepartum period, and twice a day after milking during the postpartum period using a walkthrough scale (DeLaval). Body condition score was evaluated visually on d -45 ± 3 , -21 ± 3 , 3, 23 ± 3 , 35 ± 3 , 65 ± 3 , and 90 ± 3 relative to calving using a 1 to 5 scale (Ferguson et al., 1994). Energy balance (EBAL) of prepartum and postpartum cows was estimated according to NRC (2001). Prepartum EBAL

considered the net energy intake and the net energy requirements for maintenance and pregnancy. The latter was adjusted to gestational age and BW of the calf at birth. Postpartum EBAL considered net energy intake and the net energy requirements for maintenance and lactation.

Blood Sampling and Metabolites Analyses

Blood samples were collected from the coccygeal blood vessels into polyethylene terephthalate tubes specific for TM analysis (plastic blood collection tubes for trace element testing; serum clot activator, BD Vacutainer) on d -21 ± 3 , -10 ± 1 , and -3 ± 1 before the expected calving date, and at 0, 3, 7, 10, 14, 23 ± 3 , and 65 ± 3 DIM. Samples were centrifuged at $2,860 \times g$ for 15 min at 4°C for separation of serum, which was transferred to 2-mL tubes and stored at -20°C until laboratory analysis. Serum metabolites were analyzed for a subgroup of 254 cows (STM = 123 and OTM = 131) selected randomly within those with a complete set of samples using an autochemistry analyzer (Cobas 6000 c501, Roche Diagnostics) by the University of Guelph Animal Health Laboratory (Guelph, ON, Canada). Cholesterol and glucose assays (Roche Diagnostics GmbH), nonesterified fatty acids (NEFA), and BHB assays (Randox Laboratories, Canada Ltd.) had a limit of quantification of 0.1 mmol/L. One control sample was included in each of the 15 batches of samples submitted for analysis and the CV for cholesterol, glucose, NEFA, and BHB were 3.3, 3.6, 20.4, and 6.0%, respectively.

Feeding Behavior and Rumination Activity

Feeding behavior was monitored by the automated feed bins (Insentec B.V.) that record the amount of feed consumed per visit and the start and end time of each visit. Individual feeding bouts were combined and separated into meals using a meal criterion (i.e., the minimum duration of time between meals). Three meal criteria were defined for each cow: prepartum period, postpartum period when cows were fed in individual bins, and postpartum period when cows were fed in shared bins. Meal criteria were calculated according to DeVries et al. (2003). The intervals of feed visits were $\log_{10}$ -transformed, and the frequency of the intervals was fit in a normal distribution using a software package (MIX 3.1.3; MacDonald and Green, 1988). Each meal criterion was determined as the point where the interval proportions were approximately equal. If the interval between 2 visits to the feed bin was smaller than the meal criterion, the 2 visits were considered within the same meal. If the interval between 2 visits

exceeded the meal criterion, visits were classified as different meals. Interval between meals was calculated as the average interval between feeding events that exceeded the meal criterion, and the number of meals per day was termed meal frequency (meals/d). Dry matter intake was divided by meal frequency to calculate meal size (kg of DMI/meal). Feeding time (min/d) was determined by the sum of time the cow had the head inside the feed bin. Dry matter intake was divided by feeding time to calculate feeding rate (kg of DM/min). Daily meal time was defined by the sum of feeding and nonfeeding time of each meal. Daily meal time was divided by meal frequency to calculate meal length (min/meal).

Rumination activity was monitored using an electronic monitoring system (Hr-TAG-LD, SCR Engineers Ltd.). Rumination data loggers attached to a nylon collar were fitted to each cow at 28 ± 3 d before the expected calving date and remained until 30 ± 3 DIM. The system contained a radio frequency reader, allowing for continuous data collection. Data were stored in 2-h intervals and extracted from the monitoring system weekly. The 12 daily data points were summed to obtain the total minutes of rumination per day. Daily sums were divided by the daily DMI to obtain the minutes of rumination per kilogram of DMI.

Ruminal Fluid Collection and Chemical Analysis

Ruminal fluid samples were collected from the first 235 cows enrolled in the experiment (STM = 119 and OTM = 116) between 2 and 3 h after feed delivery, at -21 ± 3 d before the expected calving date, and at 23 ± 3 and 65 ± 3 DIM, using an ororuminal sampling device attached to a glass vial and equipped with a pump, as described by Geishauser (1993). The sampling device and the glass vial were cleaned before each sample collection, using an initial wash with water at 70°C , then rinsed with 70% alcohol, followed by another rinse with water at 70°C . Ruminal fluid pH was measured immediately after collection using a glass electrode pH meter (Accumet AB150pH Benchtop Meter, Fisher Scientific) that was calibrated daily. Samples were snap-frozen in nitrogen liquid immediately after sample collection. Cows with an incomplete set of samples ($n = 16$) were excluded from posterior analyses.

The concentrations of VFA of 657 samples (219 cows; 3 samples per cow) were analyzed by gas chromatography at the Agricultural Genomics and Proteomics Laboratory of the University of Alberta. Briefly, samples were thawed and centrifuged at $21,800 \times g$ at 4°C for 5 min. The supernatant was combined with 25% of phosphoric acid (4:1, vol:vol) and centrifuged at $21,800 \times g$ at 4°C

for 5 min. One milliliter of the mixture was mixed with 200 μL of internal standard solution, composed of 25% phosphoric acid and isocaproic acid (4-methyl-valeric-acid). Samples were incubated overnight at -20°C then centrifuged at $19,000 \times g$ at 4°C for 5 min, and transferred to a GC vial, and the VFA were quantified using a gas chromatograph (GC-2010, Shimadzu) equipped with a capillary column (Stabilwax-DA 30 m, Shimadzu). Samples were run at an initial column temperature of 90°C , which was increased by $10^\circ\text{C}/\text{min}$ to 170°C and held constant for 2 min. A mixture of VFA containing acetic, propionic, isobutyric, butyric, isovaleric, valeric, and caproic acids was used as an external standard. Total VFA was calculated as the sum of each VFA analyzed. Molar proportions of each VFA were calculated as the proportion of the total VFA in the sample.

Sample Size Calculation and Statistical Analyses

Sample size calculations were performed in the WinPepi program version 11.65 (Abramson, 2011). A minimum sample size of 120 cows per treatment was calculated to allow for an 80% probability of detecting, at 5% significance level, the following differences between treatments: 1.5 kg of 3.5% ECM yield per day ($\sigma = 4.7$); 18 percentage units in the proportion of cows presenting diseases and metabolic problems (70 vs. 52%); and 14 d in median time to pregnancy (43 vs. 29) with a follow-up period of 100 d (56 to 156 d postpartum). Health and reproduction data are reported in companion papers (B. Mion, L. Ogilvie, B. Van Winters, J. F. W. Spricigo, B. W. McBride, S. J. LeBlanc, M. A. Steele, and E. S. Ribeiro).

Data were analyzed by ANOVA with mixed linear regression models using the GLIMMIX procedure of SAS on Demand (SAS Institute Inc.). Prepartum and postpartum data were analyzed separately. Statistical models included the fixed effects of treatment, parity, the interaction between treatment and parity, and period. Period was defined as 1 to 5, each representing 3-mo intervals of enrollment. For repeated measures, the fixed effects of time (either day or week), interaction between time and treatment, interaction between time and parity, and interaction between time, treatment, and parity, and the random effect of cow nested within treatment were added to the model. The covariance structure (autoregressive 1, compound symmetry, Toeplitz, unstructured, and spatial power covariance) that resulted in the lowest Bayesian information criterion was selected. For every model, residuals were tested for normality and homogeneity of variance, and data were lognormal transformed when needed. Daily

records of DMI, BW, milk yield, ECM, EBAL, feeding behavior measurements, and rumination activity were summarized weekly for statistical analyses.

For all analyses, differences with $P \leq 0.05$ were considered significant and those with $P > 0.05$ and ≤ 0.10 were declared tendencies.

RESULTS

Descriptive Statistics

A total of 282 cows were enrolled in the study, but 9 were excluded before parturition: 6 cows (4 STM and 2 OTM) failed to learn to eat exclusively from their assigned feed bins, 1 cow (STM) received the wrong feeding treatment for a period of time, 1 cow (OTM) had an accident that resulted in a broken leg, and 1 cow (OTM) was removed for teaching purposes. Data collected from the remaining 273 cows (136 STM and 137 OTM) were used for the analysis in this study. The proportion of heifers enrolled (STM: 50 heifers and 86 cows; OTM: 50 heifers and 87 cows) and the number of lactations adult cows had before enrollment (STM = 2.02 ± 0.13 vs. OTM = 1.94 ± 0.13 ; $P = 0.63$) were similar between treatment groups. Body weight (STM = 743 ± 6.5 vs. OTM = 740 ± 6.6 kg; $P = 0.74$), BCS at enrollment (STM = 3.65 ± 0.03 vs. OTM = 3.60 ± 0.03 ; $P = 0.17$), and gestation length (STM = 276.9 ± 0.43 vs. OTM = 276.8 ± 0.42 d; $P = 0.85$) were also similar between treatment groups. For cows, milk production in the previous lactation (STM = $10,622 \pm 204$ vs. OTM = $10,595 \pm 204$ kg; $P = 0.92$), DIM at dryoff

(STM = 351 ± 6.9 vs. OTM = 348 ± 6.8 ; $P = 0.78$), and length of dry period (STM = 49.8 ± 0.9 vs. OTM = 48.7 ± 0.9 d; $P = 0.38$) did not differ between treatment groups. For heifers, age at first calving was also similar between groups (STM = 24.2 ± 0.2 vs. OTM = 24.1 ± 0.2 mo; $P = 0.88$). The number of days receiving treatments during the prepartum period (STM = 42.7 ± 0.4 vs. OTM = 42.3 ± 0.4 d; $P = 0.58$) and the number of days receiving feeding treatments in a 1:1 cow to feed bin ratio during the postpartum period (STM = 38.7 ± 0.69 vs. OTM = 37.1 ± 0.65 d; $P = 0.10$) did not differ between groups. Concentrations of TM in the 4 TMR were close to target levels and not different between treatments, except for the concentration of Mn in the postpartum TMR, which was 3 mg/kg higher ($P = 0.03$) in STM than in OTM (Table 2).

Prepartum Feeding Behavior, Rumination, and Performance

Body weight and BCS did not differ ($P > 0.05$) between treatments (Table 3; Figure 1A). Cows assigned to OTM tended to eat more during the prepartum period than cows in STM, both as kg of DMI per d ($P = 0.08$) and as % of BW ($P = 0.06$; Table 3; Figure 1B, C). These tendencies in DMI were mainly explained by least squares means differences in the last week before calving (kg of DMI per d: STM = 11.27 vs. OTM = 11.84; DMI as % of BW: STM = 1.449 vs. OTM = 1.524). Cows receiving OTM had a more positive EBAL ($P = 0.05$) than cows receiving STM (Table 3; Figure 1D). There was a tendency ($P = 0.09$) for cows supplemented with OTM to have longer feeding time in the prepartum period than cows supplemented with STM (Table 3). No differences were observed in feeding rate, meal frequency, meal length, meal size, interval between meals, and daily meal time during the prepartum period. An interaction ($P = 0.04$) between treatment and time was observed for rumination activity in the last 3 wk before calving (Table 3). In wk -3 only, cows supplemented with STM tended ($P = 0.06$) to spend more time ruminating than cows supplemented with OTM (Figure 1E). Furthermore, cows receiving STM spent more time ruminating per kilogram of DMI ($P = 0.01$) than cows receiving OTM during the last 3 wk prepartum (Table 3; Figure 1F).

Postpartum Feeding Behavior, Rumination, and Performance

Differences in postpartum BW were not detected between treatments (Table 4; Figure 2A). There was a tendency ($P = 0.09$) for an interaction between parity and treatment on postpartum BCS, in which pri-

Table 2. Concentration of trace minerals in dry cow and lactating cow TMR of inorganic salts (STM) and organic (OTM) treatment groups evaluated in monthly samples by inductively coupled plasma mass spectrometry

Item	Treatment ¹		SEM	<i>P</i> -value
	STM	OTM		
Prepartum, mg/kg of DM				
Co	0.37	0.38	0.02	0.55
Cu	12.44	12.70	0.40	0.65
Mn	57.18	56.75	1.51	0.84
Se	0.45	0.41	0.02	0.23
Zn	49.22	53.06	1.99	0.19
Postpartum, mg/kg of DM				
Co	0.29	0.29	0.01	0.84
Cu	13.40	12.91	0.22	0.14
Mn	47.63	44.61	0.94	0.03
Se	0.36	0.37	0.01	0.47
Zn	59.68	57.92	1.17	0.30

¹Cows assigned to the inorganic salts group (STM) received supplementation of Co, Cu, Mn, and Zn sulfates and Na selenite. Cows assigned to the organic group (OTM) received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).

Table 3. Prepartum BW, BCS, feeding behavior, energy balance (EBAL), and rumination activity of cows supplemented with inorganic salts (STM) or organic (OTM) trace minerals

Item	Treatment, ¹ LSM $\pm$ SEM		P-value of interest ²		
	STM	OTM	TRT	TRT $\times$ parity	TRT $\times$ time
BW, kg	769.9 $\pm$ 6.0	765.0 $\pm$ 6.0	0.56	0.86	0.07
BCS, 1–5 scale	3.56 $\pm$ 0.02	3.56 $\pm$ 0.02	0.96	0.91	0.07
DMI, kg/d	12.87 $\pm$ 0.16	13.26 $\pm$ 0.16	0.08	0.21	0.74
DMI, % of BW	1.67 $\pm$ 0.02	1.73 $\pm$ 0.02	0.06	0.33	0.44
EBAL, Mcal/d	3.55 $\pm$ 0.24	4.21 $\pm$ 0.24	0.05	0.29	0.67
Rumination, min/d	478 $\pm$ 8.3	463 $\pm$ 8.3	0.20	0.34	0.04
Rumination, min/kg of DMI	40.1 $\pm$ 0.8	37.5 $\pm$ 0.8	0.01	0.64	0.97
Feeding time, min/d	188.2 $\pm$ 3.9	197.4 $\pm$ 3.9	0.09	0.62	0.24
Feeding rate, kg of DMI/min	0.08 $\pm$ 0.002	0.08 $\pm$ 0.002	0.21	0.78	0.57
Meal frequency, meals/d	7.1 $\pm$ 0.2	7.3 $\pm$ 0.2	0.62	0.93	0.33
Meal length, min/meal	44.2 $\pm$ 2.1	45.0 $\pm$ 2.0	0.76	0.12	0.62
Meal size, DMI/meal	2.3 $\pm$ 0.1	2.3 $\pm$ 0.1	0.99	0.55	0.59
Interval between meals, min	181.1 $\pm$ 4.8	177.4 $\pm$ 4.6	0.58	0.66	0.41
Daily meal time, min/d	265.7 $\pm$ 5.9	277.0 $\pm$ 5.8	0.17	0.15	0.18

¹Cows assigned to the inorganic salts group (STM, $n = 136$) received supplementation of Co, Cu, Mn, and Zn sulfates and Na selenite. Cows assigned to the organic group (OTM, $n = 137$) received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech). Data include weekly averages for the last 6 wk of gestation except for rumination activity, which includes the last 3 wk of gestation only.

²Probability values for independent variables of interest: TRT = effect of treatment (STM vs. OTM); TRT $\times$ time = interaction between treatment and time (prepartum week); TRT $\times$ parity = interaction between treatment and parity (multiparous or primiparous).

miparous cows supplemented with OTM had greater BCS than primiparous cows supplemented with STM (Table 4). Differences in BCS were not detected between treatments in multiparous cows. For DMI, there was an interaction ($P = 0.04$) between treatment and parity, in which multiparous cows supplemented with OTM had greater DMI ($P = 0.05$) than multiparous cows supplemented with STM, especially in wk 1, 3, 4, and 5 postpartum (Table 4; Figure 2B). No difference existed between treatments in DMI postpartum for primiparous cows (Table 4; Figure 2B). Similarly, when considering DMI as % BW, an interaction ($P < 0.01$) between treatment and parity was observed. Multiparous cows receiving OTM had greater ($P = 0.01$) intake than multiparous cows receiving STM (Table 4; Figure 2C), whereas primiparous cows receiving OTM tended ($P = 0.10$) to have reduced intake compared with primiparous cows receiving STM, especially on wk 13 and 14 postpartum (Table 4; Figure 2C). The difference in DMI of multiparous cows was more evident in the first 5 wk of lactation when, on average, DMI was 1 kg/d or 5.2% greater in cows supplemented with OTM than in those supplemented with STM (Figure 2C).

Postpartum feeding time, feeding rate, meal length, and daily meal time did not differ between treatments. Nevertheless, cows supplemented with OTM tended to eat more often ($P = 0.06$) and to have smaller meal size ($P = 0.07$) than cows supplemented with STM (Table 4). We found an effect ($P = 0.03$) of treatment

on the interval between meals, which was shorter for OTM compared with STM (140.1 vs. 129.3 min; Table 4). In the first 3 wk after calving, cows supplemented with OTM spent fewer minutes ruminating per day and fewer minutes ruminating per kilogram of DMI than cows supplemented with STM (Table 4; Figure 3).

A tendency for an interaction ($P = 0.06$) between parity and treatment was observed for milk yield in the first 14 wk of lactation (Table 4; Figure 2D). Although no treatment differences were observed in multiparous cows, primiparous cows supplemented with OTM produced less ($P = 0.03$) milk than primiparous cows supplemented with STM (Table 4; Figure 2D). Similar results were observed for ECM, in which differences between treatments were observed in primiparous but not in multiparous cows (Table 4; Figure 2E). Postpartum cows supplemented with OTM tended to have a less negative EBAL than cows supplemented with STM (Table 4; Figure 2F).

The first 3 milk tests were performed on average at 21, 51, and 81 DIM (Table 5). Protein yield and fat percentage on test days were not affected by treatment (Table 5). Cows supplemented with OTM had higher ($P = 0.03$) percentage of protein (STM = 3.11 vs. OTM = 3.17%) and tended ($P = 0.10$) to produce less fat (STM = 1.59 vs. OTM = 1.54 kg). There was an interaction between treatment and parity ($P = 0.05$) for protein yield, because multiparous supplemented with OTM had numerically higher protein yield (**STM-Mult** = 1.39 ± 0.02 vs. **OTM-Mult**

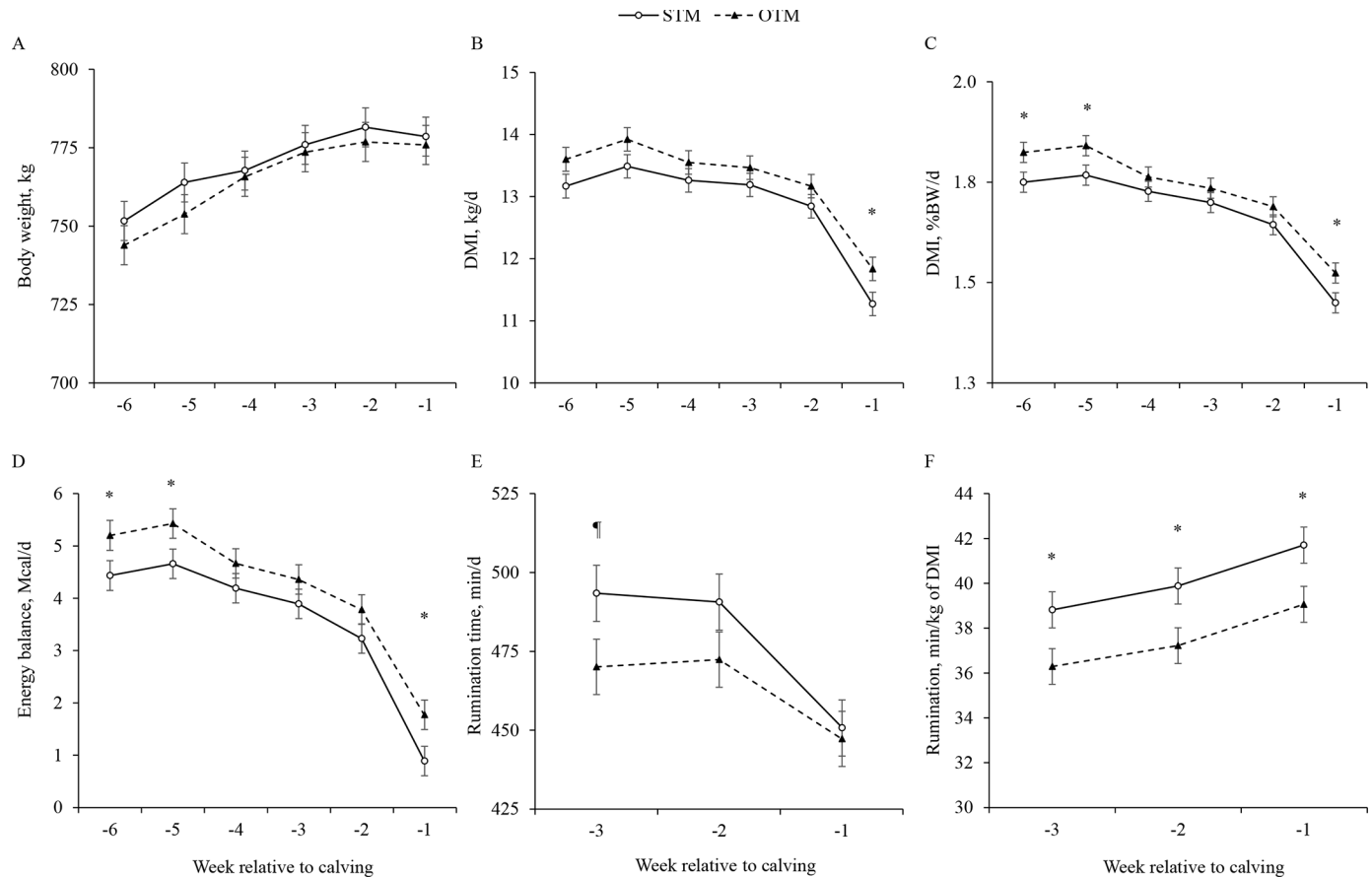


Figure 1. Prepartum BW (A), DMI (B and C), energy balance (D), and rumination activity (E and F) of cows supplemented with inorganic salts (STM) or organic (OTM) trace minerals. Data points and error bars represent the LSM and SEM, respectively. Within a week, pairwise differences between treatment ($P \leq 0.05$) and tendencies for pairwise differences between treatment ($P > 0.05$ and ≤ 0.10) are represented by * and ■, respectively.

= 1.42 ± 0.02 kg, $P = 0.25$) and primiparous supplemented with OTM had numerically lower protein yield (STM-Prim = 1.03 ± 0.02 vs. OTM-Prim = 0.97 ± 0.02 kg, $P = 0.11$) than their counterparts supplemented with STM.

Serum Metabolites

Cows supplemented with OTM had a lower concentration of NEFA (STM = 0.45 vs. OTM = 0.40 ± 0.02 mmol/L; $P < 0.01$; Figure 4) and tended to have lower concentrations of cholesterol (STM = 2.66 vs. OTM = 2.55 ± 0.04 mmol/L; $P = 0.09$) in serum, compared with cows supplemented with STM. Concentration of glucose (STM = 3.7 vs. OTM = 3.7 ± 0.04 mmol/L; $P = 0.97$) and BHB (STM = 596 vs. OTM = 578 ± 13 μ mol/L; $P = 0.97$) in serum did not differ between treatments. No interactions were present between treatment and time, or between treatment and parity, for any of the metabolites evaluated.

Ruminal Fluid pH and Volatile Fatty Acids

On d -21 , treatment had no effect on ruminal fluid pH and VFA concentration (Table 6). There was an interaction ($P = 0.04$) between treatment and parity on total VFA concentration (Table 6). Although no treatment differences were observed in primiparous cows (STM-Prim = 89.46 ± 2.30 vs. OTM-Prim = 93.37 ± 2.21 μ mol/mL, $P = 0.22$), multiparous cows supplemented with OTM tended ($P = 0.08$) to have reduced concentration of total VFA in ruminal fluid than multiparous cows supplemented with STM (STM-Mult = 92.32 ± 1.80 vs. OTM-Mult = 87.92 ± 1.78 μ mol/mL).

At 23 DIM, cows supplemented with OTM had lower ($P = 0.04$) ruminal pH than cows supplemented with STM (Table 6). In addition, cows supplemented with OTM tended ($P = 0.10$) to have greater concentration of total VFA and had a lesser ($P = 0.04$) molar proportion of acetate than cows supplemented with STM. Molar proportion of propionate, butyrate, isovalerate,

Table 4. Postpartum BW, BCS, feeding behavior, milk production, energy balance (EBAL), and rumination activity of cows supplemented with inorganic salts (STM) or organic (OTM) trace minerals¹

Item	Treatment by parity category, LSM $\pm$ SEM				<i>P</i> -value of interest ²		
	Primiparous		Multiparous		TRT	TRT $\times$ parity	TRT $\times$ time
	STM	OTM	STM	OTM			
BW, kg	606.7 $\pm$ 7.7 ^b	611.4 $\pm$ 7.7 ^b	730.4 $\pm$ 6.0 ^a	721.2 $\pm$ 6.1 ^a	0.75	0.31	0.90
BCS, 1–5 scale	3.24 $\pm$ 0.04 ^b	3.34 $\pm$ 0.04 ^a	3.15 $\pm$ 0.03 ^{bc}	3.14 $\pm$ 0.03 ^c	0.17	0.09	0.37
DMI, kg/d	19.10 $\pm$ 0.28 ^c	18.67 $\pm$ 0.28 ^c	24.10 $\pm$ 0.22 ^b	24.72 $\pm$ 0.22 ^a	0.71	0.04	0.86
DMI, % of BW	3.15 $\pm$ 0.04 ^c	3.06 $\pm$ 0.04 ^c	3.33 $\pm$ 0.03 ^b	3.45 $\pm$ 0.03 ^a	0.79	<0.01	0.72
Rumination, min/d	457 $\pm$ 11.6 ^a	425 $\pm$ 11.3 ^b	474 $\pm$ 8.5 ^a	465 $\pm$ 8.8 ^a	0.04	0.24	0.18
Rumination, min/kg of DMI	34.4 $\pm$ 1.0 ^a	33.1 $\pm$ 1.0 ^a	27.7 $\pm$ 0.7 ^b	24.8 $\pm$ 0.8 ^c	0.01	0.32	0.66
Feeding time, min/d	189.5 $\pm$ 5.8	182.8 $\pm$ 5.6	192.5 $\pm$ 4.5	194.8 $\pm$ 4.3	0.67	0.37	0.46
Feeding rate, kg of DMI/min	0.11 $\pm$ 0.004	0.11 $\pm$ 0.004	0.14 $\pm$ 0.004	0.14 $\pm$ 0.004	0.66	0.76	0.51
Meal frequency, meals/d	10.0 $\pm$ 0.4	10.5 $\pm$ 0.4	8.9 $\pm$ 0.3	9.5 $\pm$ 0.3	0.06	0.87	0.14
Meal length, min/meal	28.9 $\pm$ 1.4	28.0 $\pm$ 1.3	26.4 $\pm$ 1.6	25.0 $\pm$ 1.5	0.43	0.84	0.57
Meal size, DMI/meal	2.1 $\pm$ 0.1	1.9 $\pm$ 0.1	3.0 $\pm$ 0.1	2.9 $\pm$ 0.1	0.07	0.74	0.77
Interval between meals, min	134.9 $\pm$ 5.3 ^{ab}	122.7 $\pm$ 5.4 ^b	145.4 $\pm$ 4.1 ^a	135.9 $\pm$ 4.0 ^a	0.03	0.72	0.58
Daily meal time, min/d	255.0 $\pm$ 10.5	254.4 $\pm$ 10.1	243.6 $\pm$ 7.9	249.4 $\pm$ 7.7	0.77	0.72	0.48
Milk yield, kg/d	31.9 $\pm$ 0.7 ^b	29.8 $\pm$ 0.7 ^c	44.1 $\pm$ 0.5 ^a	44.2 $\pm$ 0.5 ^a	0.10	0.06	0.39
ECM, kg/d	34.1 $\pm$ 0.7 ^b	31.8 $\pm$ 0.7 ^c	47.7 $\pm$ 0.6 ^a	47.7 $\pm$ 0.6 ^a	0.07	0.07	0.58
EBAL, Mcal/d	–1.02 $\pm$ 0.45 ^a	–0.26 $\pm$ 0.44 ^a	–2.88 $\pm$ 0.35 ^b	–2.35 $\pm$ 0.34 ^b	0.08	0.89	0.91

^{a–c}Least squares means in the same row with different superscripts differ ($P \leq 0.05$) in pairwise comparisons.

¹Cows assigned to the inorganic salts group (STM, $n = 136$) received supplementation of Co, Cu, Mn, and Zn sulfates and Na selenite. Cows assigned to the organic group (OTM, $n = 137$) received supplementation of Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech). Data include weekly averages for the first 14 wk of lactation except for rumination activity, which includes the first 3 wk of lactation only.

²Probability values for independent variables of interest: TRT = effect of treatment (STM vs. OTM); TRT $\times$ parity = interaction between treatment and parity (multiparous or primiparous); TRT $\times$ time = interaction between treatment and time (postpartum week).

valerate, isobutyrate, and caproate did not differ between treatments. When VFA was evaluated as concentration, there were tendencies for cows supplemented with OTM to have greater concentration of propionate (STM = 23.60 ± 0.69 vs. OTM = 25.31 ± 0.67 $\mu\text{mol/mL}$, $P = 0.07$), butyrate (STM = 10.72 ± 0.31 vs. OTM = 11.47 ± 0.30 $\mu\text{mol/mL}$, $P = 0.08$), and valerate (STM = 1.56 ± 0.05 vs. OTM = 1.69 ± 0.05 $\mu\text{mol/mL}$, $P = 0.06$), compared with cows supplemented with STM. No differences in ruminal pH, total VFA, and molar proportion of VFA were observed between treatments at 65 DIM.

DISCUSSION

Feed intake is commonly depressed around the time of calving, which limits intake of energy and essential nutrients (e.g., TM and vitamins) during a period of increased nutrient demands for fetal growth and the onset of lactation (Drackley et al., 2006). Inadequate energy and nutrient supplies during the transition period can lead to metabolic imbalances, oxidative stress, and impair immune function (Sordillo, 2016). In fact, the magnitude of feed intake depression around the time of parturition is associated with poorer postpartum health and performance of dairy cows (Hammon et al., 2006;

Huzzey et al., 2007; Pérez-Báez et al., 2019). Identification of managerial and nutritional strategies that minimize the depression in intake around parturition is important, and the present results indicate that OTM supplementation could contribute modestly to greater feed intake during the transition period, consequently improving EBAL and reducing the levels of circulating NEFA. For multiparous cows, DMI in the first 5 wk after calving was 1 kg/d, or 5.2% greater, in OTM groups than in STM group. Regardless of parity, concentration of NEFA in serum from 3 to 10 of DIM was on average 0.1 mmol/L, or 17% higher, in STM cows than in OTM cows. Extensive mobilization of body reserves and high concentration of NEFA in the early postpartum period have been associated with increased risk of clinical health problems and culling, reduced milk production, and reproductive performance (Ospina et al., 2010a,b; Chapinal et al., 2011).

Researchers have previously reported changes in feed intake according to sources of supplementary TM. In contrast to our results, Osorio et al. (2016) reported a reduction in prepartum DMI and a tendency for a greater postpartum DMI. However, in the study performed by Osorio et al. (2016), only partial replacement of STM (Co, Cu, Mn, and Zn) by OTM sources through oral boluses was performed during the transi-

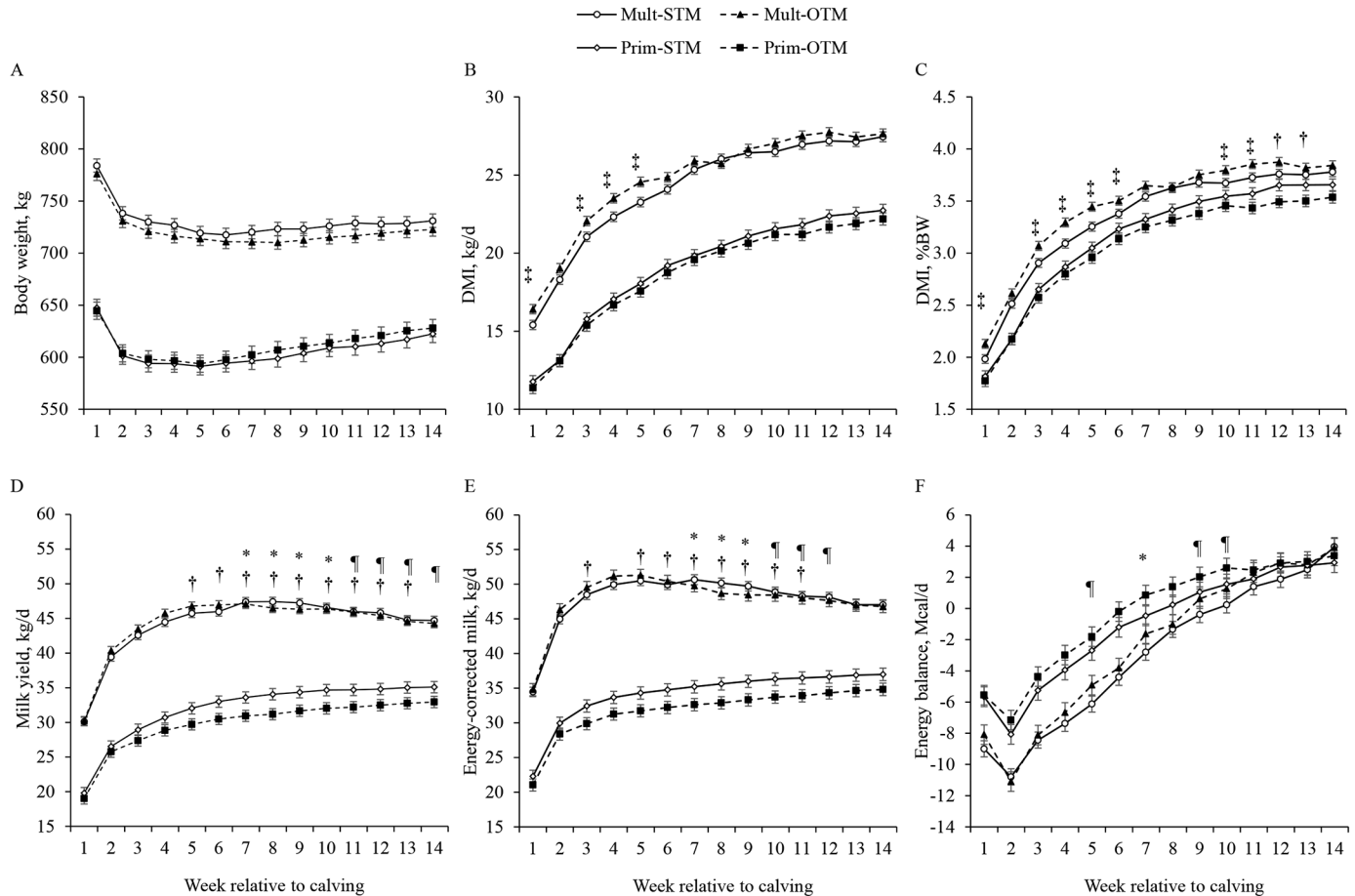


Figure 2. Postpartum BW (A), DMI (B and C), milk production (D and E), and energy balance (F) of cows supplemented with salts (STM) or organic (OTM) trace minerals. Data points and error bars represent the LSM and SEM, respectively. Within a week, pairwise differences ($P \leq 0.05$) are represented as follows: *STM vs. OTM; †primiparous STM vs. primiparous OTM; and ‡multiparous STM vs. multiparous OTM. In addition, tendencies for pairwise differences between treatment ($P > 0.05$ and ≤ 0.10) are represented by §.

tion period, whereas in our study we performed total replacement of STM by OTM sources in both pre- and postpartum TMRs. Chen et al. (2020) reported that supplementation of Zn-methionine given to prepartum cows increased DMI linearly, with increasing inclusion levels from 0 to 60 mg/kg of DM. Changes in feed intake are normally accompanied by changes in feeding behavior. In the present study, the observed increase in DMI in the OTM group was accompanied by a tendency to greater feeding time in the prepartum period, and tendencies for more frequent meals and shorter intervals between meals in the postpartum period. Johnston and DeVries (2018) combined multiple data sets in which DMI and feed behavior were closely monitored and reported that feeding time is positively associated with DMI. In contrast to our results, Pino and Heinrichs (2016) reported that heifers supplemented with OTM (Cu, Mn, Se, and Zn) had shorter eating time than

heifers supplemented with STM; however, differences in DMI were not observed in that study.

It is unclear why supplementary TM source would affect feed intake and feeding behavior in dairy cows. Nonetheless, changes in rumen microbial function are a possible explanation for the observed outcomes, although limited information is available to substantiate this possibility (Pino and Heinrichs, 2016; Kljak et al., 2017). Rumen microbes require TM for proper function and the excess of some TM have a toxic effect (Hubbert et al., 1958; Martinez and Church, 1970; Durand and Kawashima, 1980). Supplementary sources with different biochemistry properties, solubility, and stability in ruminal fluid could affect TM utilization by microbes and, consequently, their speed or efficiency of fermentation of feed (Genther and Hansen, 2015). Guimaraes et al. (2020) reported an increase in the digestibility of NDF and ADF in beef steers when fed a dairy diet

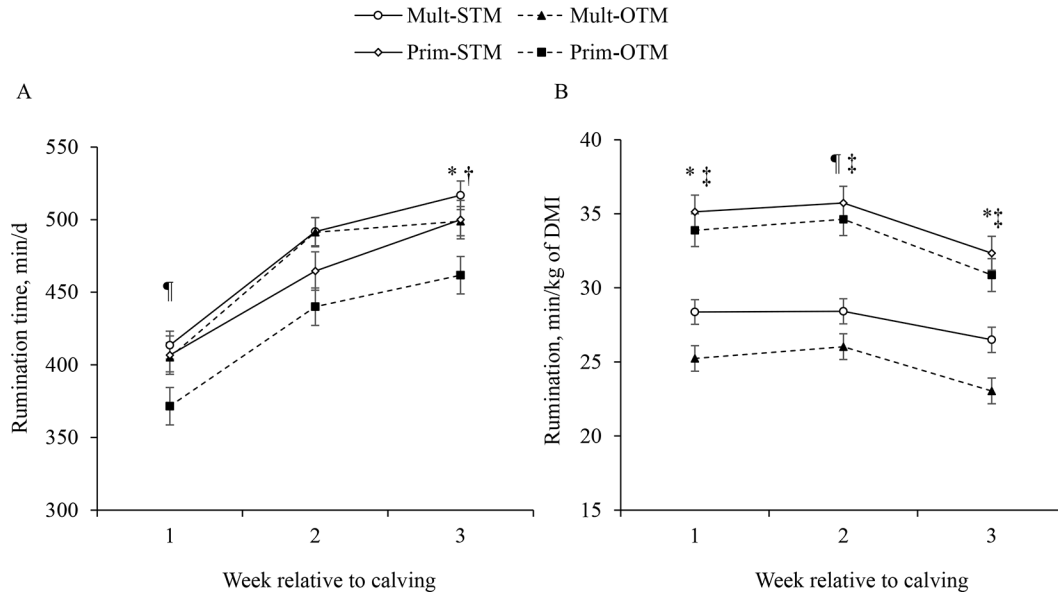


Figure 3. Postpartum rumination activity of primiparous (Prim) and multiparous (Mult) cows supplemented with salts (STM) or organic (OTM) trace minerals. Panel A shows minutes of rumination activity per day, and panel B shows minutes of rumination per kilogram of DMI. Data points and error bars represent the LSM and SEM, respectively. Within a week, pairwise differences ($P \leq 0.05$) are represented as follows: *STM vs. OTM; †primiparous STM vs. primiparous OTM; and ‡multiparous STM vs. multiparous OTM. In addition, tendencies for pairwise differences between treatment ($P > 0.05$ and ≤ 0.10) are represented by ¶.

Table 5. Milk yield and composition at the first 3 DHI tests of postpartum cows supplemented with inorganic salts (STM) or organic (OTM) trace minerals

Item	Treatment, ¹ LSM $\pm$ SEM		<i>P</i> -value of interest ²		
	STM	OTM	TRT	TRT $\times$ parity	TRT $\times$ test
DIM			0.88	0.13	0.88
First test	21 $\pm$ 0.9	21 $\pm$ 0.9			
Second test	51 $\pm$ 0.9	51 $\pm$ 0.9			
Third test	82 $\pm$ 0.9	81 $\pm$ 0.9			
Protein, %			0.03	0.44	0.98
First test	3.21 $\pm$ 0.02	3.26 $\pm$ 0.02			
Second test	3.00 $\pm$ 0.02 ^b	3.06 $\pm$ 0.02 ^a			
Third test	3.11 $\pm$ 0.02 ^b	3.18 $\pm$ 0.02 ^a			
Protein yield, kg			0.58	0.05	0.53
First test	1.16 $\pm$ 0.02	1.17 $\pm$ 0.02			
Second test	1.21 $\pm$ 0.02	1.19 $\pm$ 0.02			
Third test	1.26 $\pm$ 0.02	1.23 $\pm$ 0.02			
Fat, %			0.96	0.84	0.90
First test	4.42 $\pm$ 0.07	4.39 $\pm$ 0.07			
Second test	3.92 $\pm$ 0.05	3.93 $\pm$ 0.05			
Third test	3.87 $\pm$ 0.05	3.88 $\pm$ 0.05			
Fat yield, kg			0.10	0.26	0.91
First test	1.61 $\pm$ 0.03	1.57 $\pm$ 0.03			
Second test	1.59 $\pm$ 0.03	1.53 $\pm$ 0.03			
Third test	1.56 $\pm$ 0.03	1.50 $\pm$ 0.03			

^{a,b}Least squares means in the same row with different superscripts differ ($P \leq 0.05$) in pairwise comparisons.

¹Cows assigned to the inorganic salts group (STM, $n = 136$) received supplementation of Co, Cu, Mn, and Zn sulfates and Na selenite. Cows assigned to the organic group (OTM, $n = 137$) received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).

²Probability values for independent variables of interest: TRT = effect of treatment (STM vs. OTM); TRT $\times$ parity = interaction between treatment and parity (multiparous or primiparous); TRT $\times$ test = interaction between treatment and test number (first, second, or third).

supplemented with OTM (Cu, Mn, and Zn) or HTM, compared with the same diet supplemented with STM. Moreover, in a study comparing STM and HTM (Cu, Mn, and Zn) in mid-lactation Holstein cows, Miller et al. (2020b) reported greater DMI and total-tract digestibility of amylase NDF on an organic matter basis (aNDFom) for the HTM group. The reported advantages of OTM and HTM in comparison to STM on rumen function could be a result of better TM incorporation by rumen microbes (Galbraith et al., 2016; Pino and Heinrichs, 2016) or reduced toxicity caused by excessive dissociation of TM in ruminal fluid (Genther and Hansen, 2015; Faulkner and Weiss, 2017) when OTM or HTM are used.

It is noteworthy that rumination activity (total and per kg of DM) during the transition period was reduced in cows supplemented with OTM in our study, despite their greater feed intake. If fiber digestibility was in fact enhanced, it is possible that OTM increased rumen outflow and, consequently, reduced rumination activity, but this hypothesis needs to be confirmed. Rumination time and DMI are often described as being positively correlated. Johnston and DeVries (2018) predicted an increase of 0.2 kg of daily DMI for every hour of increase in rumination time above the mean. However, the positive association between the 2 variables is not consistent in the literature. Schirrmann et al. (2012)

reported no association between DMI and rumination time in prepartum cows.

Rumination time is highly dependent on dietary NDF and physically effective NDF (Allen, 1997). In our study, both treatment groups received the same diet, except for the source of TM. Therefore, differences in rumination cannot be attributed to differences in NDF. Interestingly, the source of TM in the diet affected the VFA profile on d 23 after calving, suggesting a slight improvement of rumen fermentation toward the end of the transition period. Cows supplemented with OTM had slightly lower ruminal pH, a tendency to have greater concentration of total VFA in ruminal fluid, reduced molar proportion of acetate, and a tendency for greater concentrations of propionate, butyrate, and valerate. In a study performed by Wang et al. (2009), it was demonstrated that supplementing lactating cows with selenized yeast resulted in greater concentration of total VFA and molar proportion of propionate and reduced the ratio of acetate to propionate. Pino and Heinrichs (2016) demonstrated that heifers supplemented with OTM sources (Cu, Mn, Se, and Zn) had reduced ruminal pH and greater concentration of butyrate.

Although the supplementation with OTM sources had beneficial effects on DMI and VFA production during transition to lactation, milk yield and ECM were not improved by OTM supplementation. In fact, primiparous cows receiving OTM produced approximately 2 kg/d less milk and ECM than primiparous cows receiving STM. The results of other experiments on the effect of the TM source on milk production are inconsistent, and the variable results seem mostly related to the diversity of TM sources and experimental designs. Siciliano-Jones et al. (2008) reported a tendency for greater milk and ECM yields when partially replacing STM (Co, Cu, Mn, and Zn) by OTM during the transition period. Similarly, Osorio et al. (2016) reported beneficial effects on milk production when partially replacing STM (Co, Cu, Mn, and Zn) by OTM sources through oral boluses during the transition period. In contrast, Daniel et al. (2020) described no differences in DMI and yields of milk, fat, and protein when replacing 30% of supplementary STM or HTM (Cu, Mn, and Zn) with OTM in the diet of mid-lactation cows. For complete replacements, Yasui et al. (2019) reported no difference in milk production, milk components, and DMI when mid-lactation multiparous cows were fed STM (Cu, Mn, and Zn) or OTM for 6 wk. In contrast, Nocek et al. (2006) reported greater milk, fat, and protein yields and reduced SCC when replacing STM (Co, Cu, Mn, and Zn) by OTM for 2 lactation cycles.

A meta-analysis by Rabiee et al. (2010) evaluated 20 experiments, including peer-reviewed publications,

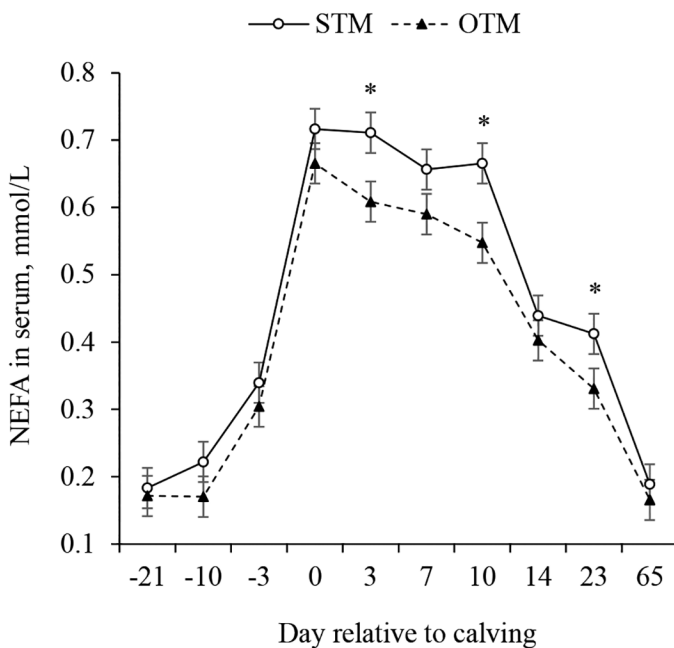


Figure 4. Concentration of nonesterified fatty acids (NEFA) in serum of transition cows supplemented with salts (STM) or organic (OTM) trace minerals. Data points and error bars represent the LSM and SEM, respectively. Within a week, pairwise differences between treatment ($P \leq 0.05$) are represented by *.

Table 6. Ruminal fluid pH and VFA concentrations on d -21 , 23 , and 65 ± 3 relative to calving according to supplementary trace mineral treatment

Item	Treatment, ¹ LSM $\pm$ SEM		<i>P</i> -value of interest ²		
	STM	OTM	TRT	Parity	TRT $\times$ parity
d -21 ± 3					
pH	6.82 $\pm$ 0.03	6.82 $\pm$ 0.03	0.92	0.93	0.26
VFA, mol/100 mol of VFA					
Acetate	67.85 $\pm$ 0.26	67.77 $\pm$ 0.25	0.80	<0.01	0.28
Propionate	19.31 $\pm$ 0.20	19.36 $\pm$ 0.19	0.85	<0.01	0.35
Butyrate	9.00 $\pm$ 0.15	9.13 $\pm$ 0.14	0.50	0.20	0.71
Isovalerate	1.47 $\pm$ 0.03	1.42 $\pm$ 0.03	0.15	0.47	0.75
Valerate	1.01 $\pm$ 0.03	1.01 $\pm$ 0.03	0.86	0.48	0.36
Isobutyrate	0.95 $\pm$ 0.02	0.94 $\pm$ 0.02	0.54	0.22	0.64
Caproate	0.40 $\pm$ 0.02	0.37 $\pm$ 0.02	0.39	0.14	0.27
Total VFA, μ mol/mL	90.89 $\pm$ 1.48	90.64 $\pm$ 1.43	0.90	0.52	0.04
d 23 ± 3					
pH	6.61 $\pm$ 0.03	6.53 $\pm$ 0.03	0.04	0.57	0.35
VFA, mol/100 mol of VFA					
Acetate	62.06 $\pm$ 0.44	60.85 $\pm$ 0.43	0.04	<0.01	0.85
Propionate	22.55 $\pm$ 0.46	23.48 $\pm$ 0.45	0.14	<0.01	0.91
Butyrate	10.31 $\pm$ 0.22	10.51 $\pm$ 0.21	0.49	0.13	0.61
Isovalerate	1.92 $\pm$ 0.06	1.96 $\pm$ 0.06	0.63	<0.01	0.37
Valerate	1.49 $\pm$ 0.04	1.57 $\pm$ 0.04	0.12	0.17	0.16
Isobutyrate	0.97 $\pm$ 0.02	0.95 $\pm$ 0.02	0.38	0.97	0.93
Caproate	0.69 $\pm$ 0.04	0.68 $\pm$ 0.04	0.79	0.22	0.34
Total VFA, μ mol/mL	103.66 $\pm$ 1.95	108.06 $\pm$ 1.89	0.10	0.47	0.79
d 65 ± 3					
pH	6.57 $\pm$ 0.03	6.53 $\pm$ 0.03	0.34	0.21	0.59
VFA mol/100 mol of VFA					
Acetate	62.46 $\pm$ 0.40	61.89 $\pm$ 0.39	0.29	0.83	0.44
Propionate	21.96 $\pm$ 0.41	22.39 $\pm$ 0.40	0.44	0.43	0.16
Butyrate	2.95 $\pm$ 0.07	2.87 $\pm$ 0.06	0.36	0.60	0.14
Isovalerate	1.81 $\pm$ 0.04	1.86 $\pm$ 0.04	0.35	0.72	0.47
Valerate	1.61 $\pm$ 0.03	1.65 $\pm$ 0.03	0.34	0.73	0.93
Isobutyrate	1.00 $\pm$ 0.02	1.00 $\pm$ 0.02	0.89	<0.01	0.17
Caproate	0.81 $\pm$ 0.04	0.80 $\pm$ 0.04	0.84	0.54	0.35
Total VFA μ mol/mL	109.99 $\pm$ 1.92	111.49 $\pm$ 1.87	0.57	0.57	0.65

¹Cows assigned to the inorganic salts group (STM, $n = 136$) received supplementation of Co, Cu, Mn, and Zn sulfates and Na selenite. Cows assigned to the organic group (OTM, $n = 137$) received supplementation with Co, Cu, Mn, and Zn proteinates (Bioplex, Alltech) and selenized yeast (Sel-Plex, Alltech).

²Probability values for independent variables of interest: TRT = effect of treatment (STM vs. OTM); parity = effect of parity (multiparous vs. primiparous); TRT $\times$ parity = interaction between treatment and parity.

published abstracts, and technical bulletins, which evaluated the inclusion of OTM products from a single company in dairy cow diets. Studies included a variety of experimental designs with multiple TM inclusion levels, including top-up, partial, or total replacement of STM sources and variable duration of supplementation. Those researchers concluded that OTM supplementation is expected to increase milk production by 0.9 kg/d. However, the data were highly heterogeneous and influenced greatly by a single large study with large positive effects on milk production (Nocek et al., 2006). The meta-analysis did not find differences in percentages of protein and fat with the inclusion of OTM in the diets. In the present study, the percentage of protein in milk was slightly greater in OTM, which might be related to reported effects of TM on tissue protein synthesis (Otter et al., 1989; Kimball et al., 1995) or in

microbial protein synthesis (Durand and Kawashima, 1980).

It is unclear why lower milk production was associated with complete replacement of STM by OTM in primiparous cows. Considering that no differences in DMI between treatments were observed during lactation in primiparous cows, the difference in milk production is likely a result of changes in diet digestibility or changes in energy and nutrient partitioning. The latter seems more likely because primiparous cows supplemented with OTM had higher postpartum BCS than primiparous cows supplemented with STM. Moreover, no major differences in rumen function outcomes measured in this study were observed between primiparous cows of the 2 treatments. A biological mechanism for the potential changes in energy and nutrient partitioning is unclear, but it could be related to oxidative balance

and insulin sensitivity. Chalmeh et al. (2021) reported that supplementation of Se yeast enhanced insulin sensitivity in transition dairy cows, compared with an unsupplemented control. Parental supplementation of folic acid and vitamin B12, a vitamin that requires Co for endogenous synthesis, also resulted in greater insulin sensitivity in transition dairy cows (Girard et al., 2019). Although TM or antioxidants might be important to insulin sensitivity and potentially to energy partitioning, there are no reports that those effects would be exclusive to primiparous cows. Thus, additional research is needed to evaluate potential differences in TM requirements, utilization, and physiological responses to different supplementary TM sources and dietary inclusion levels in primiparous cows. Most studies on TM nutrition in dairy cattle have been performed in multiparous cows, and it is not uncommon for first lactation cows to present different responses to treatments than older cows, considering that their physiology and metabolism differ substantially (Wathes et al., 2007).

CONCLUSIONS

Complete replacement of supplementary STM by OTM in pre- and postpartum diets of dairy cows resulted in changes in rumen function, feeding behavior, energy metabolism, and milk yield, with some of these responses dependent of parity status. During the prepartum period, DMI tended to be greater and EBAL was greater with OTM supplementation, irrespective of parity. During the postpartum period, supplementation with OTM increased the percentage of protein in milk, reduced the interval between meals and the concentration of NEFA in serum, and tended to increase EBAL. In addition, OTM improved early postpartum DMI in multiparous cows, whereas it decreased milk production and increased BCS in primiparous cows. Regarding rumen function outcomes, OTM supplementation reduced rumination activity during the transition period and, by the end of transition period, slightly reduced pH and the molar proportion of acetate, and tended to increase the concentration of total VFA in ruminal fluid. Different responses to TM supplementation forms between primiparous and multiparous cows suggest differences in requirements or utilization of TM according to parity, thus warranting further research on supplementation strategies tailored to specific parity, especially for primiparous cows.

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